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Research article

Design of optimal cascade control approach for LFM of interconnected power system

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ABSTRACT

In the present era, due to increasing power demand and complex power system structures having various load disturbances, a load frequency management (LFM) scheme is indispensable to provide uninterrupted power to consumers. This research deals with a fractional-order proportional derivative - (one + fractional order integrator) (FOPD-(1+FOI)) cascade controller as a novel control structure to ameliorate the execution of automatic generation control (AGC) for the LFM of interconnected power system (PS). The implementation of this controller is uncomplicated, and it joins the output of the FOPD controller to (1+FOI) controller, where area control error and power error are considered in the outer and inner feedback control loops, respectively. A maiden attempt of a wild horse optimizer-assisted FOPD-(1+FOI) cascade controller for AGC of considered interconnected PS has been performed in this work. To benchmark the proposed control scheme, two areas reheat thermal PS with GDB and GRC nonlinearities is chosen as the test bench. A vivid comparative analysis of six state-of-the-art control techniques is performed, and the results reveal the potency of the presented control approach. Eigenvalues-based stability assessment of interconnected PS in conjunction with the proposed controller is also performed. Finally, for the real PS implementation of the presented control architecture a new england IEEE 39 test bus is considered and analyzed.

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1. Introduction

1.1. Background

Due to racing power demand in today's world, including PS size spurt, complexity, and interconnected PS network, it is a very challenging task to provide secure, uninterrupted, and stable power via proper power flow management [1]. The power demand is continuously fluctuating in nature, resulting in momentary divergence in tie-line flow and frequency of the entire PS [2–4]. Various control areas are connected to a PS network. From the electrical utility point of view, for stable and reliable operation, power generated in each control area should meet the demands, and total losses occurred; as well as tie-line and deviation of frequency from reference should fall within permissible close to zero limits. Consequently, noteworthy and rapid expansion has been seen in the LFM area because it is impossible to maintain the balance among demands–generations without supplementary control action [5,6]. In real-time, AGC is the control action that stabilizes the demand–generation balance and maintains the frequency deviations of the entire PS within permissible limits. Particularly, LFM forces the synchronous generator units to adjust their output to fulfill the demands. Taking note of the above-mentioned discourse, an enhanced and adequate control scheme is an unavoidable demand for the LFM of PSs. Moreover, for the purpose of engineering commissioning, the practical potential and suitability of an LFM control scheme in a real-time execution is a much-needed feature [7–9].

1.2. Literature review

From the comprehensive literature survey, it is pointed out that there are enormous highly appreciated works in the area of LFM present in the last decades. Every article is striving for further enhancement. In Ref. [2], PIDD2 controller based on the internal model control and worst-case plant selection scheme is proposed for the LFM of time-delayed interconnected PS. A new performance index based on a hybrid peak area for the AGC of interconnected PS is briefed in [3]. In this, the authors have performed a comparative analysis of the proposed performance index with the existing indices. A fuzzy logic (FL) -based adaptive LFM of renewable energy shared standalone microgrid is demonstrated in [5]. In this, a fuzzy observer is used to extract the power

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Abbreviations

ABC	Artificial bee colony
ACO	Ant colony optimization
BWOA	Black widow optimization algorithm
COA	Coyote optimization algorithm
DE	Differential evolution
DSA	Dragonfly search algorithm
FA	Firefly algorithm
GA	Genetic algorithm
GDB	Governor dead band
GOA	Grasshopper optimization algorithm
GRC	Generation rate constraint
GSA	Gravitational search algorithm
hBFOA	Hybrid bacterial foraging optimization algorithm
HHO	Haris hawk optimization
IAE	Integral of absolute error
ISE	Integral of squared error
ITAE	Integral of time multiplied absolute error
MBA	Mine blast algorithm
MGWO	Modified grey wolf optimization
MVO	Multi-verse optimization
OS	Overshoot
PDF	Proportional–derivative with filter
PS	Power system
PSO	Particle swarm optimization
QOHS	Quasi-oppositional harmony search
SBL	Stability boundary locus
SSA	Salp swarm algorithm
TSA	Tunicate swarm algorithm
US	Undershoot
WHO	Wild horse optimizer

from wind turbine generator, and an improved incremental conductance scheme is used to harvest maximum power from the solar. A FL-based fractional variant of the PID controller is used to investigate the AGC performance of interconnected PS [7]. A few articles that are recently published employing PI/PID-based controllers are (hBFOA-PSO) [10], BFOA [11], FA [12], lozi-map based chaotic algorithm [13], MGWO [14], ACO [15], QOHS [16], GA [17], DSA [18], DE [19], symbiotic organisms search [20], ABC [21], GWO tuned PI/PID controllers [22], and jaya algorithm assisted PID with low pass filter [23]. To palliate control mechanism, other enhanced control techniques based on H-infinity [24], sliding mode control [25], fractional calculus theory [26], model predictive control [27,28], FL approach [29], nonlinear disturbance observer [30], ANN [31], and ANFIS [32] are presented in the relevant field. A double-stage PI-(1+PD) controller is designed via a GOA scheme for the LFM of wind-ocean-based maritime PS in [33]. A comparative analysis among PID, PIDF, and PI-(1+PD) tuned via GA, PSO, FA, and GOA is performed. Design consideration of the cascade controller for the frequency regulation of multi-area PS based on COA is briefed in [34]. The simulation results revealed the proposed cascade PDF-PI controller's superior and robust dynamic performance under parametric uncertainties. The maximum frequency deviations under the worst condition are not exceeding 0.4% for the proposed control scheme. Online LFM in wind integrated PSs via modified jaya optimization is described in [35]. The performance assessment of SSA tuned intelligent fuzzy 1PD-PI controller for the AGC of renewable rich multi-area PS is presented in Ref. [36]. In this, author has designed a cascade control architecture for the frequency regulation with the stability evaluation of the proposed technique. A frequency deviation mitigation technique via COA tuned fuzzy controller under wind integrated PS is briefed in [37]. A frequency control mechanism of distributed energy resources based on coordinated strategies is revealed in [38]. In this, small signal dynamic models for renewable energy sources are introduced in standalone microgrid. Design of active disturbance rejection based LFM of time delayed PS is demonstrated in [39]. Fraction order controller is designed for the AGC of multi area PS in Refs. [40,41].

The PS works under various uncertainty conditions due to stochastic load demands, inherent nonlinearities, communication delays, and continuous variation in its structure. Though the intelligent algorithms tuned conventional controllers, as mentioned above, show good performance, their performance may greatly affect in regards to longer settling time and larger oscillations. Moreover, the output of the traditional controllers optimized for a specific point will worsen with changing operating conditions of the system. The fractional and cascade concept of controller design is much appreciated for its enhanced disturbance rejection capability before it is transmitted to other sections and as underlined by researchers. A cascade control structure has the potency to provide enhanced performance to the traditional controller [42]. In this regard, MGWO tuned PI-PD cascade controller [14], BWOA-tuned PIDF-(1+I) controller [43], bat algorithm-driven PD-PID cascade controller [44], DE-based multi-stage PDF plus (1+PI) controller [45] are briefed for LFM of various PSs.

From the above-mentioned vivid literature review, the critical observations are; (1) the implementation of the metaheuristic approach on the fuzzy-logic, ANN and ANFIS, makes the system very complex, (2) designed controllers in the literature are just standard feedback control-based approach, and (3) there is always a scope to implement robust, advanced, and recently invented metaheuristic approach for the LFM of the power system.

1.3. Motivation and contributions

The system responses of these cascade controllers show interesting control outcomes in the area of LFM. However, as per the best knowledge of the authors' results obtained in these literature have some space for improvement while considering the time domain indices such as OS, US, rise time, and settling time, as well as system performance in terms of errors such as *IAE*, *ITAE*, *ISE*, *andITSE*. The presented study revisits LFM by orienting a novel factional and cascade controller named FOPD-(1+FOI), whose benchmarking is not yet investigated so far. In view of the inspected literature, various nature-inspired optimizations are implemented in LFM studies for the purpose of proper gain tuning and to obtain the possible outcome from the controllers. WHO, a recently invented metaheuristic algorithm proposed by Iraj Naruei and Farshid Keynia (2021) [46] has proved its efficacy over GA, PSO, GWO, MVO, TSA, LFD, SSA, GSA, and HHO in various problems such as [46]; process synthesis and design, heat exchange design, two-reactor problem, tension/compression spring design, step-cone pully, three bar truss design, rolling-element bearing, and gas transmission compressor design, so it is considered for the LFM of the considered interconnected PS. A vivid comparative analysis of the proposed LFM controller design in terms of time-domain indices and performance of controller in terms of errors is performed with other metaheuristics like; GWO, MBA, MGWO, and recently invented BWOA [47], and other control structures like; FOPID, PI-PD, and PID. Moreover, a stability evaluation of the proposed LFM scheme is also performed in this work. For the stability evaluation, the

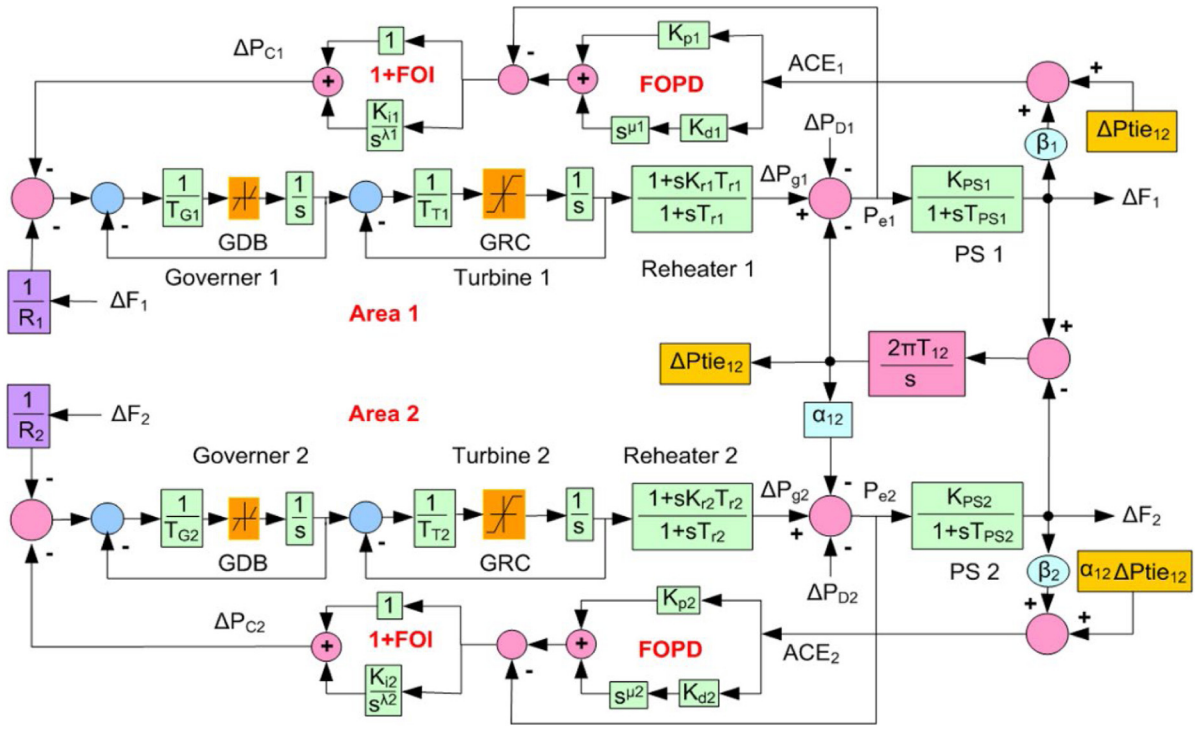


Fig. 1. Two-areas reheat thermal PS with FOPD-(1+FOI) cascade control.

fractional orders are approximated into their respective full-order terms via the SBL method [48]. In comparison with the existing literature, the significant contributions of the work are revealed as:

(1) Design of WHO tuned FOPD-(1+FOI) fractional cascade controller for LFM of two areas PS and its superiority has been established over designed FOPID, PI-PD, and PID controllers.
(2) Performance assessment of WHO is verified over MBA, GWO, and BWOA concerning minimum fitness value, cost-effectiveness, and convergence rate.
(3) Benchmarking of the proposed controller's robustness concerning parametric uncertainties and load disturbances.
(4) Eigenvalues-based stability assessment of the proposed LFM scheme.
(5) For real power system validation, the proposed LFM technique is implemented on New England IEEE 39 test bus.

The rest of the part is documented as; Section 2 deals with the studied PS model, and Section 3 describes the vivid design consideration of the proposed controller. A vivid description of WHO is presented in Section 4, results and discussion with stability evaluation and real power system implementation are revealed in Section 5. The final concluding remark is given in Section 6.

2. Power system model under investigation

A two-area interconnected reheat thermal PS is investigated and revealed in Fig. 1, and each PS control area has a load of 2000 MW. The reheat thermal PS with GDB and GRC has been comprehensively captivated in the AGC literature [2,3,6,8]. The generation speed is restricted due to the practical moderation of thermal and mechanical activities shown by GRC. On the other hand, GDB is a measure of continuous speed variation in which there is no deviation in the position of the valve. In order to conduct a real-time investigation on AGC in the presence of nonlinearities, GDB and GRC must be considered. In Fig. 1, the various components are as: area control error (*ACE*), governor regulation (R), frequency biasing (β), controller output (ΔP_c), turbine time constant (TC) (T_T), governor TC (T_G), reheater TC (T_r), reheater gain (K_r), change in load (ΔP_D), power error P_e, turbine power (ΔP_g), PS gain (K_{PS}), PS TC (T_{PS}) and capacity ratio of area is (α_{12}). Moreover, ΔF_i (frequency variation), $\Delta Ptie_{ij}$ (tie-line power variation), and T_{ij} (synchronizing coefficient).

The detailed parameters of the considered two-area reheat thermal PS are provided in Appendix. The generalized dynamics of multiarea PS in the form of a transfer function (TF) are described as follows [2]:

$$\text{Governor } TF = G_{G,i}(s) = \frac{1}{(1+sT_{Gi})} \tag{1}$$

$$\text{Reheated turbine } TF = G_{T,i}(s) = \frac{1}{(1+sT_{Ti})}\frac{(1+sK_{ri}T_{ri})}{(1+sT_{ri})} \tag{2}$$

$$\text{PS } TF = G_{PS,i}(s) = \frac{K_{PSi}}{(1+sT_{PSi})} \tag{3}$$

where $i = 1, 2 \ldots N$.

3. Proposed controller design

3.1. Fractional order (FO) calculus

The generalized form of integer order calculus is the FO calculus that converts the integer-order differential and integral operators to the FO. ${}_A\mathrm{Q}_t^{\overline{\alpha}}$ is the fractional operator responsible for the transform. Based on the sign of $\overline{\alpha}$, the differentiation or integration operation is decided. The modeling of the fractional operator is as follows [40]:

$${}_A\mathrm{Q}_t^{\overline{\alpha}} = \begin{cases} \frac{d^{\overline{\alpha}}}{dt^{\overline{\alpha}}}, & \overline{\alpha} > 1 \\ 1, & \overline{\alpha} = 0 \\ \int_{\overline{\alpha}}^{t} (dt)^{\overline{\alpha}}, & \overline{\alpha} < 1 \end{cases} \tag{4}$$

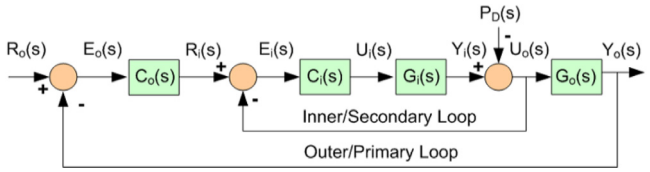


Fig. 2. Structure of proposed cascade controller.

There is various literature available for the definitions of the FO functions. A few definitions are the Grunwald–Letnikov (G–L) definition, Riemann–Liouville (R–L) definition, and the Caputo definition [40].

The most used definition is R–L, which is as follows [40]:

$$ {}_{A}Q_t^{\overline{\alpha}} = \frac{1}{\Gamma(k-\overline{\alpha})}\frac{d^k}{dt^k}\int_{\overline{\alpha}}^{t}\frac{x(\tau)}{(t-\tau)^{1-(k-\overline{\alpha})}}d\tau \tag{5} $$

where, $\Gamma(\cdot)$ is the Euler's gamma function, and k is an integer $k-1 < \overline{\alpha} \leq k$. Before using the FO controller in real or simulation applications, it is required the approximation of FO differentiators and integrators with corresponding integer-order counterparts. Moreover, the FO differentiators and integrators are infinite-dimensional in nature, so before putting them into practice, must be band limited. To band limit them, various approximation schemes are available such as Matsuda approximation, Carlson approximation, Oustaloup's recursive approximation, and Crone approximation [41]. In this proposed work, Oustaloup's recursive approximation is employed to realize FO controllers within the frequency range of $(\omega_l = 0.001, \omega_u = 1000)$ rad/s. Using this, FO differentiator and integrator are approximated via recursive distribution of zeros and poles as:

$$ s^{\overline{\alpha}} = M\prod_{k=1}^{K}\frac{1+\left(\frac{s}{\omega_{z,k}}\right)}{1+\left(\frac{s}{\omega_{p,k}}\right)} \tag{6} $$

Frequencies of zeros and poles are:

$$ \begin{cases} \omega_{z1} = \omega_l\sqrt{\eta} \\ \omega_{z,k} = \omega_{z,k}\varepsilon,\ k = 1,2,3,\ldots K \\ \omega_{z,k+1} = \omega_{p,k}\sqrt{\eta},\ k = 1,2,3,\ldots K-1 \\ \varepsilon = \left(\frac{\omega_u}{\omega_l}\right)^{\overline{\alpha}/K} \\ \eta = \left(\frac{\omega_u}{\omega_l}\right)^{(1-\overline{\alpha})/K} \end{cases} \tag{7} $$

In this work, the authors have utilized the fractional-order modeling and control (FOMCON) toolbox for the designing of the proposed FO controller.

3.2. Cascade controller

The key feature of the cascade controller is to mitigate the impact of disturbance propagation through another part of the system since it gives an extra feedback path, i.e., sensor for restricting and reducing disturbance quickly before the final output [34]. It consists of two loops, as revealed in Fig. 2.

The outer loop is considered the primary loop that carries $Y_o(s)$ as the controlled output. Its gain $G_o(s)$ is affected by the power demand $P_D(s)$ and expressed as:

$$ Y_o(s) = G_o(s)\,U_o(s) = G_o(s)\,(Y_i(s) - P_D(s)) \tag{8} $$

where, $U_o(s)$ is the input signal to $G_o(s)$, which is equivalent to the inner loop's processed output $Y_i(s)$. The outer loop's controller is $C_o(s)$, which will supervise $Y_o(s)$ to receive a reference signal $R_o(s)$. The inner loop is considered the secondary loop and will process the inner loop's gain $G_i(s)$ and formulated as:

$$ Y_i(s) = G_i(s)\,U_i(s) \tag{9} $$

where, $U_i(s)$ is the input signal to the $G_i(s)$ and $C_i(s)$ is the controller for the inner loop. Its objective is to reduce the inner process gain deviations impact. The complete transfer function for the cascade control is revealed as follows:

$$ Y_o(s) = \left[\frac{G_o(s)\,G_i(s)\,C_o(s)\,C_i(s)}{1+G_o(s)\,C_o(s)+G_o(s)\,G_i(s)\,C_o(s)\,C_i(s)}\right]R_o(s) - \left[\frac{G_o(s)}{1+G_o(s)\,C_o(s)+G_o(s)\,G_i(s)\,C_o(s)\,C_i(s)}\right]P_D(s) \tag{10} $$

In this work, the proposed cascade controller is FOPD-(1+FOI), where the primary/outer loop controller $C_o(s)$ is FOPD, and the inner loop controller $C_i(s)$ is (1+FOI), as depicted in Fig. 1. The two controllers can be described in the generalized form as:

$$ C_{o,i}(s) = K_{pi} + K_{di}s^{\mu i} \tag{11} $$

$$ C_{i,i}(s) = 1 + \frac{K_{ii}}{s^{\lambda i}} \tag{12} $$

where $i = 1$ for area-1 and $i = 2$ for area-2, and K_{pi} is the proportional gain, K_{di} is differential gain, μi is differential fractional order, K_{ii} is integral gains, and λi is integral fractional order. Due to the cascade connection of two controllers, it obtains the property of disturbance rejection before they outspread to another system. From Fig. 1, the resulting equations for area-1 are as:

$$ \Delta F_1 = G_{PS,1}(s)P_{e1} \tag{13} $$

$$ P_{e1} = \Delta P_{g1} - \Delta Ptie_{12} - \Delta P_{D1} \tag{14} $$

$$ \Delta P_{g1} = G_{G,1}(s)G_{T,1}(s)\left(-\Delta P_{c1} - \frac{1}{R_1}\Delta F_1\right) \tag{15} $$

$$ \Delta P_{c1} = C_{i,1}(s)\left(C_{o,1}(s) * ACE_1 - P_{e1}\right) \tag{16} $$

$$ ACE_1 = \beta_1\Delta F_1 + \Delta Ptie_{12} \tag{17} $$

$$ \Delta Ptie_{12} = \frac{2\pi T_{12}}{s}(\Delta F_1 - \Delta F_2) \tag{18} $$

To establish the independent relations for area-1, considering the $\Delta F_2 = 0$ in (18), and from (13)–(18), yield (19) (see Box I). Similarly, the decoupled equation for area-2 is derived and represented by (20) (see Box I). To obtain the optimal controller performance, a total of ten parameters in both areas of PS must be optimized. WHO is chosen for the simultaneous optimization of these parameters. The optimized value of any controller depends on the appropriate objective function (J). Various types of J have been utilized in the literature as mentioned below:

$$ (\text{ISE}) = \int_0^t \left\{(\Delta F_1)^2 + (\Delta F_2)^2 + (\Delta Ptie_{12})^2\right\}dt \tag{21} $$

$$ (\text{ITSE}) = \int_0^t t\left\{(\Delta F_1)^2 + (\Delta F_2)^2 + (\Delta Ptie_{12})^2\right\}dt \tag{22} $$

$$ (\text{IAE}) = \int_0^t \{|\Delta F_1| + |\Delta F_2| + |\Delta Ptie_{12}|\}\,dt \tag{23} $$

$$ (\text{ITAE}) = \int_0^t t\,\{|\Delta F_1| + |\Delta F_2| + |\Delta Ptie_{12}|\}\,dt \tag{24} $$

where t is the simulation time, ΔF_1, ΔF_2 and ΔPtie_{12} are the responses against step load perturbation (SLP). Controlled responses with damped oscillations and less settling time are obtained while dealing with ITAE objective function, and it is implemented in LFM studies very frequently [6,33–35] compared to other variants; hence, it is considered in the presented study. The design of the controller is an obligate tuning operation, where the

$$\Delta F_1 = -\left[\frac{G_{PS,1}(s)}{1+\frac{2\pi T_{12}}{s}G_{PS,1}(s)-G_{G,1}(s)G_{T,1}(s)C_{i,1}(s)+G_{G,1}(s)G_{T,1}(s)G_{PS,1}(s)C_{i,1}(s)C_{o,1}(s)\left(\beta_1+\frac{2\pi T_{12}}{s}\right)+\frac{G_{G,1}(s)G_{T,1}(s)G_{PS,1}(s)}{R_1}}\right]\Delta P_{D1} \quad (19)$$

$$\Delta F_2 = -\left[\frac{G_{PS,2}(s)}{1+\frac{2\pi T_{12}}{s}G_{PS,2}(s)-G_{G,2}(s)G_{T,2}(s)C_{i,2}(s)+G_{G,2}(s)G_{T,2}(s)G_{PS,2}(s)C_{i,2}(s)C_{o,2}(s)\left(\beta_2+\frac{2\pi T_{12}}{s}\right)+\frac{G_{G,2}(s)G_{T,2}(s)G_{PS,2}(s)}{R_2}}\right]\Delta P_{D2} \quad (20)$$

Box I.

bounds parameter of the controller works as a constraint. Hence, lessen J incorporating to:

$$\left\{\begin{array}{c} K_{p1}^{min} \le K_{p1} \le K_{p1}^{max} \\ K_{d1}^{min} \le K_{d1} \le K_{d1}^{max} \\ K_{i1}^{min} \le K_{i1} \le K_{i1}^{max} \\ K_{p2}^{min} \le K_{p2} \le K_{p2}^{max} \\ K_{d2}^{min} \le K_{d2} \le K_{d2}^{max} \\ K_{i2}^{min} \le K_{i2} \le K_{i2}^{max} \\ \mu 1^{min} \le \mu 1 \le \mu 1^{max} \\ \mu 2^{min} \le \mu 2 \le \mu 2^{max} \\ \lambda 1^{min} \le \lambda 1 \le \lambda 1^{max} \\ \lambda 2^{min} \le \lambda 2 \le \lambda 2^{max} \end{array}\right\} \quad (25)$$

These ten parameters are obtained via running WHO to enhance the complete system performance of the proposed interconnected PS.

4. Wild horse optimizer (WHO)

The behavior of non-territorial horses is the motivation for this optimization technique which was proposed in 2021 [46]. WHO consists of five key steps; in the first step initial population is created, and horse groups are formed with the selection of leaders; mating and grazing of horses is the second step; in the third step, the groups are led by the leader (stallion); the process of exchange and selection of leaders have occurred in the fourth step, and in the final step, the best solution is obtained. For proper communication, stallions are located near the mares, and mating occurs at any time. Generally, foals are started grazing in the initial week and have less rest time as they become older. The random initial population is considered as follows:

$$(\vec{x}) = \left\{\vec{x}_1, \vec{x}_2, \vec{x}_3 \ldots\ldots\ldots\ldots\ldots, \vec{x}_n\right\} \quad (26)$$

The process of evaluation of this random population is performed via the target function. The target function is evaluated as:

$$\left(\vec{O}\right) = \{O_1, O_2, O_3 \ldots\ldots\ldots\ldots\ldots O_n\} \quad (27)$$

Firstly, the population is divided into various groups. The number of groups is $G = [n \times ps]$, where n is the number of members and ps is the percentage of stallions, and it is the control parameter in this algorithm. This population division is revealed in Fig. 3.

To obtain the grazing behavior, the stallion is placed in the center of the grazing area, and other members search around the center. The grazing behavior is simulated as:

$$\overline{X}_{i,G}^{j} = 2Z\cos(2\pi RZ) \times \left(Stallion^{j} - X_{i,G}^{j}\right) + Stallion^{j} \quad (28)$$

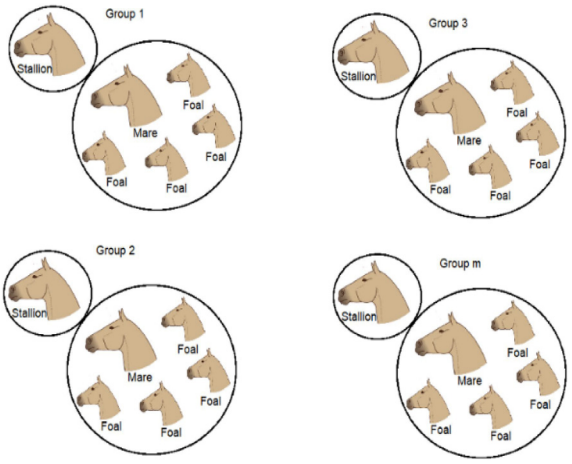


Fig. 3. Formation of horse groups in initial population [46].

where, $\overline{X}_{i,G}^{j}$ is the new place of group member while grazing, the current position of foal or mare is $X_{i,G}^{j}$, the position of the group leader is $Stallion^{j}$, R is uniform distribution in $[-2, 2]$, Z is the adaptive mechanism calculated as:

$$P = \vec{R}_1 < TDR;\ IDX = (P == 0);\ Z = R_2 \Theta IDX + \vec{R}_3 \Theta (\sim IDX) \quad (29)$$

P is a vector with values 0 and 1, $\vec{R}_1$, R_2 and $\vec{R}_3$ are random vectors with $[0, 1]$, IDX is the index of a random vector $\vec{R}_1$ satisfying $(P == 0)$, TDR is an adaptive parameter with an initial value 1 and decreased to 0 as the execution of the algorithm starts as:

$$TDR = 1 - iter \times \left(\frac{1}{maxiter}\right) \quad (30)$$

where, $iter$ and $maxiter$ are the current and maximum iterations. One of the important behavior of horses is the segregation of foals from the batch and mating them while compared with other animals. The mating process of horses is simulated as:

$$X_{G,K}^{p} = Crossover\left(X_{G,i}^{q}, X_{G,i}^{z}\right)\ i \ne j \ne k,\ p = q = end,$$
$$Crossover = Mean \quad (31)$$

In group k, the position of horse p is $X_{G,K}^{p}$ and gives it a place who have to quit groups i and j. The location of foal q is $X_{G,i}^{q}$, in group i it mated with a horse (z) with position $\left(X_{G,i}^{z}\right)$, which leaves group

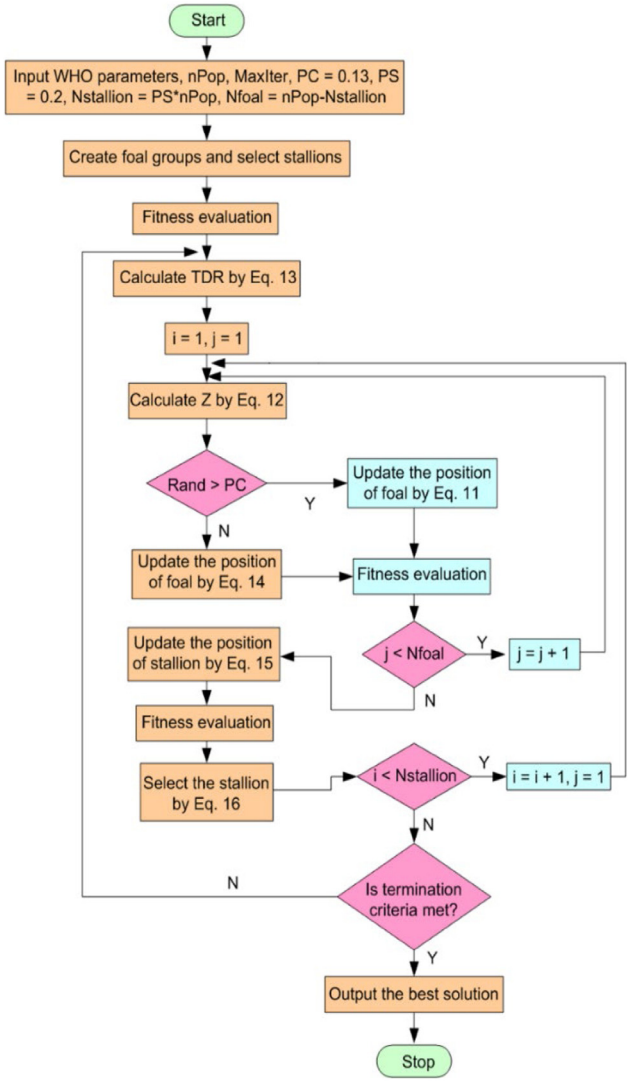


Fig. 4. Flowchart of WHO algorithm [46].

j. The group leader has the capability to lead the group to an appropriate area (water hole), and this process is simulated by:

$$\overline{Stallion_{G_i}} = \left\{ \begin{array}{l} 2Z\cos(2\pi RZ) \times \left(WH - Stallion_{G_i}\right) + WHifR_3 > 0.5 \\ 2Z\cos(2\pi RZ) \times \left(WH - Stallion_{G_i}\right) - WHifR_3 \le 0.5 \end{array} \right\} \tag{32}$$

$$\overline{Stallion_{G_i}} = \left\{ \begin{array}{c} X_{G,i} if cost\left(X_{G,i}\right) < \cos t\left(Stallion_{G_i}\right) \\ Stallion_{G_i} if cost\left(X_{G,i}\right) > \cos t\left(Stallion_{G_i}\right) \end{array} \right\} \tag{33}$$

where, next location of leader is $\overline{Stallion_{G_i}}$ of the group *i*, location of water hole is *WH*, and present location of leader is $Stallion_{G_i}$ of the group *i*. Flowchart of the WHO scheme is shown in Fig. 4.

5. Results and discussion

5.1. Test case-1 with SLP of 10% in area-1

The proposed WHO:FOPD-(1+FOI) fractional cascade controller is implemented on an interconnected reheat thermal PS. A state-of-the-art of five optimization techniques as MBA, GWO, MGWO, BWOA, and WHO, and four different types of controllers as PID, FOPID, PI-PD, FOPD-(1+FOI) for the LFM have been briefed. To benchmark the superior performance of fractional cascade controller, the convergence rates of WHO:PID, WHO:FOPID, WHO:PI-PD, and WHO:FOPD-(1+FOI) are depicted in Fig. 5(a)–(d), respectively, and analyzed in Table 1.

In the case of WHO:PID the best score is obtained after 77 iterations with a minimum *J* of 0.1; for WHO:FOPID the best score

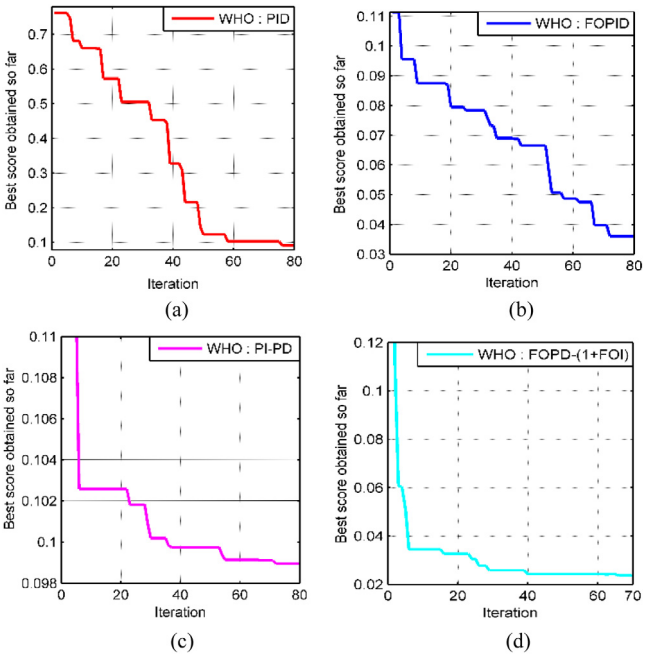


Fig. 5. Convergence of WHO tuned (a) PID, (b) FOPID, (c) PI-PD and (d) FOPD-(1+FOI).

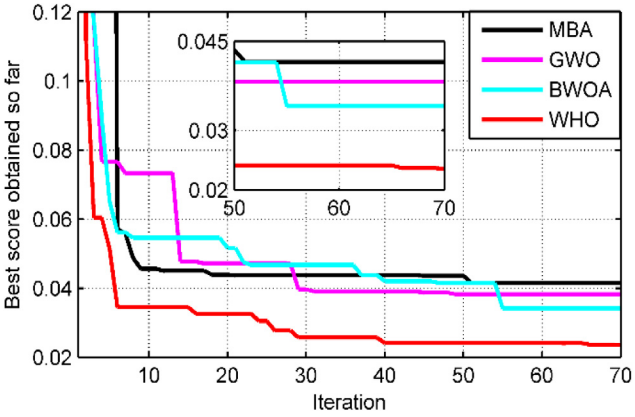


Fig. 6. Convergence rate of various metaheuristics for FOPD-(1+FOI) controller.

Table 1
Convergence analysis of all designed control schemes.

Techniques	Iterations	Minimum *J*
WHO:PID	77	0.1
WHO:FOPID	72	0.038
WHO:PI-PD	75	0.099
MBA:FOPD-(1+FOI)	53	0.0415
GWO:FOPD-(1+FOI)	49	0.0395
BWOA:FOPD-(1+FOI)	56	0.0366
Proposed	**40**	**0.0234**

is obtained after 72 iterations with a minimum J of 0.038; for WHO:PI-PD, the best score is obtained after 75 iterations with a minimum J of 0.099, while for the proposed WHO:FOPD-(1+FOI) controller the best score is obtained after 40 iterations with minimum J of 0.0234. So, from the above-mentioned analysis and results revealed in Fig. 5(a)–(d) it is concluded that the proposed fractional cascade controller is fast enough and cost-effective in comparison to full order controller, fractional-order controller, and full-order cascade controller. Fig. 6 and Table 1 have revealed the convergence of different meta-heuristics applied to FOPD-(1+FOI) controller. It is noted that the best score with faster convergence and cost-effectiveness is obtained in the case of the proposed WHO scheme with a minimum J of 0.0234 in comparison to MBA, GWO, and BWOA. An SLP of 10% is provided in area-1, and the responses as ΔF_1, ΔF_2 and $\Delta Ptie$ via the implementation of various control schemes are presented in Fig. 7(a)–(c), respectively.

The maximum OS in ΔF_1 is 0.01 Hz (GWO:PID) and minimum for the proposed WHO:FOPD-(1+FOI) controller (0.0023 Hz). The maximum OS of 2.4×10^{-3} Hz is obtained for ΔF_2 in the case of MBA:PID and minimum are noted for the proposed controller as 0.24×10^{-3}. While analyzing the US, the maximum US in ΔF_1 is 0.0097 Hz (GWO:PID), and the minimum US is 0.0029 Hz obtained for the proposed control scheme. As depicted in Fig. 7(b) and Table 2, the USs in ΔF_2 are 0.0023 Hz, 0.0028 Hz, 0.00192 Hz, 0.00125 Hz, 0.0011 Hz, and 0.0008 Hz for MBA:PID, GWO:PID, MGWO:PI-PD, WHO: FOPID, BWOA:FOPD-(1+FOI) and proposed controller respectively. Moreover, the minimum US of the order of 0.00035 is obtained for WHO:FOPD-(1+FOI) controller while dealing with the response of $\Delta Ptie$. While analyzing the settling time (t_s) of ΔF_1, the t_s of 3.3 s, 2.3 s, 2.4 s, 1.5 s, 1.3 s, and 1.3 s for MBA:PID, GWO:PID, MGWO:PI-PD, WHO: FOPID, BWOA: FOPD-(1+FOI) and proposed controller respectively is obtained. The maximum t_s are 3.55 s, and 3 s for ΔF_2 and $\Delta Ptie$ respectively for MBA:PID controller, and the minimum is noted for the proposed WHO:FOPD-(1+FOI) controller as mentioned in Table 2. Further while dealing with the system performance in terms of errors for ΔF_1, ΔF_2 and $\Delta Ptie$, the minimum of IAE, ITAE, ISE, and ITSE is noted for the proposed controller, and the maximum is noted for MBA:PID controller.

5.2. Test case-2 with SLP of 20% in area-1 and SLP of 10% in area-2

In this section, the analysis of the performance of all applied controllers for LFM is evaluated. Fig. 8(a)–(c) show the response of ΔF_1, ΔF_2 and $\Delta Ptie$ and vivid comparative benchmarking of the controller outputs is revealed in Table 3.

The maximum OS (0.11 Hz) of ΔF_1 is obtained in the case of MBA-PID, and the minimum OS (0.0065 Hz) is for the proposed controller. For ΔF_2 OSs are 0.024 Hz, 0.013 Hz, 0.0065 Hz, 0.017 Hz, 0.0013 Hz, and 0.0008 Hz for MBA:PID, GWO:PID, MGWO:PI-PD, WHO:FOPID, BWOA:FOPD-(1+FOI) and proposed controller respectively, while the USs are 0.028 Hz, 0.022 Hz, 0.018 Hz, 0.014 Hz, 0.0036 Hz, and 0.0018 Hz for the same applied controller. While dealing with the $\Delta Ptie$, the maximum OS and US are 0.009 and 0.011 for MBA:PID, and the minimum OS and US are 0.0002 and 0.0006 for the proposed controller. From Fig. 8(a)–(c) and Table 3, the $t_s(\Delta F_1)$ are 2.4 s, 2 s, 1.9 s, 1.8 s, 1.3 s, and 1.2 s, the $t_s(\Delta F_2)$ are 2.6 s, 2.5 s, 2.3 s, 2.4 s, 2.2 s, 2 s for MBA:PID, GWO:PID, MGWO:PI-PD, WHO:FOPID, BWOA:FOPD-(1+FOI), and proposed controller respectively. In the case of tie-line flow the t_s are, 3.2 s, 2.7 s, 2.4 s, 2.2 s, 2 s, 2 s for MBA:PID, GWO:PID, MGWO:PI-PD, WHO:FOPID, BWOA:FOPD-(1+FOI) and WHO:FOPD-(1+FOI) controllers respectively. While dealing with the system performance in terms of error, the minimum of IAE, ITAE, ISE, andITSE is obtained for the proposed

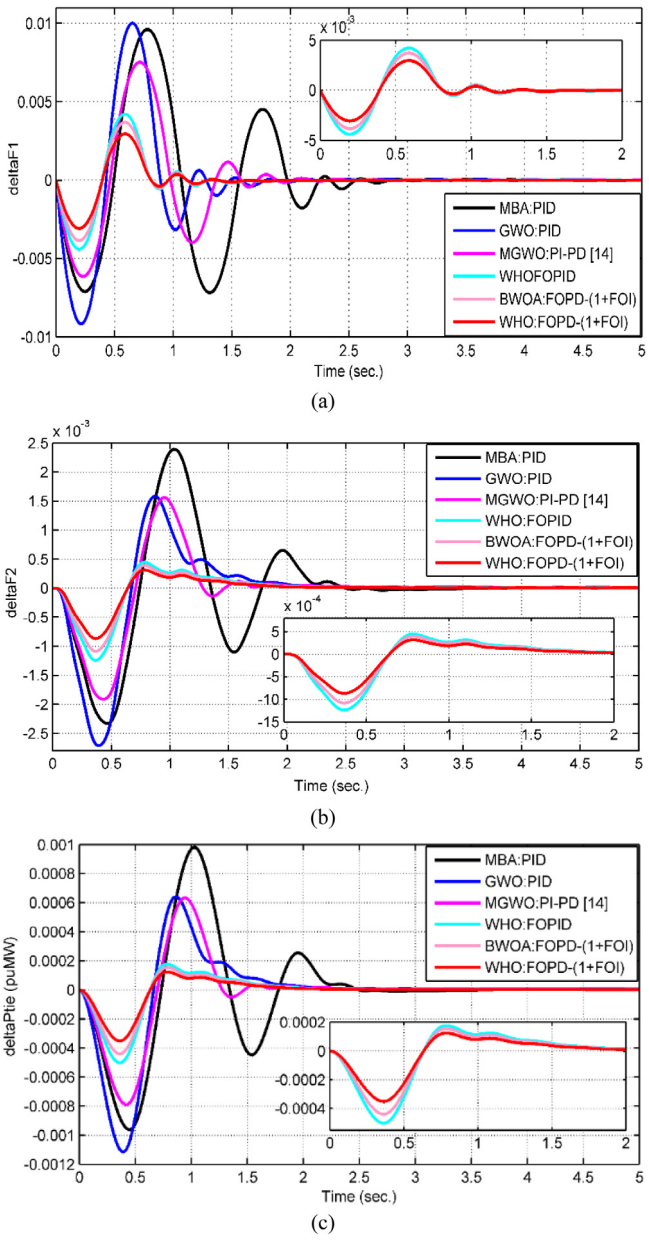


Fig. 7. Performance of controllers with SLP of 10% in area 1 (a) ΔF_1 (Hz) (b) ΔF_2 (Hz), and (c) $\Delta Ptie$.

controller, and the maximum IAE, ITAE, ISE, andITSE has obtained for MBA:PID controller for ΔF_1, ΔF_2 and $\Delta Ptie$ as tabulated in Table 3.

5.3. Robustness analysis

In this section, the robustness assessment of the proposed WHO:FOPD-(1+FOI) controller under parametric uncertainties is evaluated. In the practical system, parametric uncertainties may occur in the governor, turbine, reheater, frequency biasing, and governor regulation for both area-1 and area-2 also, there will be communication delay in the physical response of all connected elements. The above-mentioned uncertainties will greatly affect the performance of PS; hence a huge number of uncertainties are considered only in the parameters of PS of area-1 (T_{PS1}andK_{PS1}) with SLP of 13% in area-1, and the responses of the proposed controller are depicted in Fig. 9(a)–(c), and 10(a)–(c) with vivid analysis tabulated in Table 4. Firstly, the uncertainties in T_{PS1} as

Table 2
Comparative assessment of control schemes for SLP of 10% in area-1.

SLP of 10% in area-1							
Performance indices		MBA: PID	GWO: PID	MGWO: PI-PD [14]	WHO: FOPID	BWOA: FOPD-(1+FOI)	WHO: FOPD-(1+FOI)
Overshoot (OS)	ΔF_1 (Hz)	0.00950	0.0100	0.0078	0.00410	0.0038	**0.0023**
	ΔF_2 (Hz)	0.00240	0.00152	0.00151	0.00048	0.00046	**0.00024**
	$\Delta Ptie$ (puMW)	0.00100	0.00062	0.00062	0.00019	0.00017	**0.0001**
Undershoot (US)	ΔF_1 (Hz)	0.00770	0.0097	0.00610	0.0043	0.0039	**0.0029**
	ΔF_2 (Hz)	0.00230	0.0028	0.00192	0.00125	0.0011	**0.00080**
	$\Delta Ptie$ (puMW)	0.00095	0.0011	0.0008	0.0005	0.00045	**0.00035**
Rise time (t_r) (s)	ΔF_1	0.50000	0.4500	0.4700	0.4000	0.4000	**0.4000**
	ΔF_2	0.80000	0.7000	0.7500	0.6500	0.6500	**0.6500**
	$\Delta Ptie$	0.75000	0.7100	0.7300	0.6200	0.6200	**0.6200**
Settling time (t_s) (s)	ΔF_1	3.3000	2.3000	2.4000	1.5000	1.3000	**1.3000**
	ΔF_2	3.5500	2.4000	2.3000	2.0000	1.9000	**1.9000**
	$\Delta Ptie$	3.000	3.0000	2.4000	1.9000	1.9000	**1.9000**
ΔF_1	*IAE*	6.1×10^{-1}	7.8×10^{-2}	4.3×10^{-2}	4.0×10^{-2}	3.9×10^{-2}	$\mathbf{3.3\times10^{-2}}$
	ITAE	5.8×10^{-1}	6.5×10^{-2}	3.9×10^{-2}	3.6×10^{-2}	3.32×10^{-2}	$\mathbf{2.9\times10^{-2}}$
	ISE	7.6×10^{-2}	8.85×10^{-3}	2.2×10^{-3}	1.9×10^{-3}	1.7×10^{-3}	$\mathbf{1.1\times10^{-3}}$
	ITSE	8.1×10^{-3}	9.3×10^{-4}	4.1×10^{-4}	3.7×10^{-4}	3.4×10^{-4}	$\mathbf{3.1\times10^{-4}}$
ΔF_2	*IAE*	5.65×10^{-1}	7.1×10^{-2}	3.9×10^{-2}	3.8×10^{-2}	3.6×10^{-2}	$\mathbf{2.95\times10^{-2}}$
	ITAE	5.23×10^{-1}	5.8×10^{-2}	3.74×10^{-2}	3.3×10^{-2}	3.12×10^{-2}	$\mathbf{2.65\times10^{-2}}$
	ISE	6.23×10^{-2}	6.25×10^{-3}	1.9×10^{-3}	1.6×10^{-3}	1.3×10^{-3}	$\mathbf{8.21\times10^{-4}}$
	ITSE	7.43×10^{-3}	7.54×10^{-4}	3.8×10^{-4}	3.3×10^{-4}	2.3×10^{-4}	$\mathbf{2.3\times10^{-4}}$
$\Delta Ptie$	*IAE*	5.28×10^{-1}	6.85×10^{-2}	3.25×10^{-2}	2.65×10^{-2}	3.55×10^{-2}	$\mathbf{2.85\times10^{-2}}$
	ITAE	4.91×10^{-1}	5.61×10^{-2}	3.54×10^{-2}	2.81×10^{-2}	2.91×10^{-2}	$\mathbf{2.54\times10^{-2}}$
	ISE	5.96×10^{-2}	5.89×10^{-3}	1.81×10^{-3}	1.35×10^{-3}	1.25×10^{-3}	$\mathbf{7.86\times10^{-4}}$
	ITSE	7.13×10^{-3}	7.21×10^{-4}	3.6×10^{-4}	2.86×10^{-4}	2.12×10^{-4}	$\mathbf{2.23\times10^{-4}}$

Table 3
Comparative assessment of control schemes for SLP of 20% in area-1 and SLP of 10% in area-2.

SLP of 20% in area-1 and SLP of 10% in area-2							
Performance Indices		MBA: PID	GWO: PID	MGWO: PI-PD [14]	WHO: FOPID	BWOA: FOPD-(1+FOI)	WHO: FOPD-(1+FOI)
OS	ΔF_1 (Hz)	0.110	0.0800	0.0600	0.0420	0.0130	**0.0065**
	ΔF_2 (Hz)	0.024	0.0130	0.0065	0.0170	0.0013	**0.0008**
	$\Delta Ptie$ (puMW)	0.009	0.0055	0.0028	0.0015	0.0005	**0.0002**
US	ΔF_1 (Hz)	0.090	0.0750	0.0610	0.0500	0.0130	**0.0065**
	ΔF_2 (Hz)	0.028	0.0220	0.0180	0.0140	0.0036	**0.0018**
	$\Delta Ptie$ (puMW)	0.011	0.0090	0.0070	0.0057	0.0014	**0.0006**
t_r (s)	ΔF_1	0.480	0.4600	0.4200	0.3800	0.4200	**0.4200**
	ΔF_2	0.700	0.7200	0.7000	0.6500	0.7000	**0.7000**
	$\Delta Ptie$	0.700	0.6800	0.6600	0.5800	0.6600	**0.6600**
t_s (s)	ΔF_1	2.400	2.0000	1.9000	1.8000	1.3000	**1.2000**
	ΔF_2	2.600	2.5000	2.3000	2.4000	2.2000	**2.0000**
	$\Delta Ptie$	3.200	2.7000	2.4000	2.2000	2.0000	**2.0000**
ΔF_1	*IAE*	6.1×10^{-1}	7.8×10^{-2}	4.3×10^{-2}	4.0×10^{-2}	3.9×10^{-2}	$\mathbf{3.3\times10^{-2}}$
	ITAE	5.8×10^{-1}	6.5×10^{-2}	3.9×10^{-2}	3.6×10^{-2}	3.32×10^{-2}	$\mathbf{2.9\times10^{-2}}$
	ISE	7.6×10^{-2}	8.85×10^{-3}	2.2×10^{-3}	1.9×10^{-3}	1.7×10^{-3}	$\mathbf{1.1\times10^{-3}}$
	ITSE	8.1×10^{-3}	9.3×10^{-4}	4.1×10^{-4}	3.7×10^{-4}	3.4×10^{-4}	$\mathbf{3.1\times10^{-4}}$
ΔF_2	*IAE*	5.65×10^{-1}	7.1×10^{-2}	3.9×10^{-2}	3.8×10^{-2}	3.6×10^{-2}	$\mathbf{2.95\times10^{-2}}$
	ITAE	5.23×10^{-1}	5.8×10^{-2}	3.74×10^{-2}	3.3×10^{-2}	3.12×10^{-2}	$\mathbf{2.65\times10^{-2}}$
	ISE	6.23×10^{-2}	6.25×10^{-3}	1.9×10^{-3}	1.6×10^{-3}	1.3×10^{-3}	$\mathbf{8.21\times10^{-4}}$
	ITSE	7.43×10^{-3}	7.54×10^{-4}	3.8×10^{-4}	3.3×10^{-4}	2.3×10^{-4}	$\mathbf{2.3\times10^{-4}}$
$\Delta Ptie$	*IAE*	5.28×10^{-1}	6.85×10^{-2}	3.25×10^{-2}	2.65×10^{-2}	3.55×10^{-2}	$\mathbf{2.85\times10^{-2}}$
	ITAE	4.91×10^{-1}	5.61×10^{-2}	3.54×10^{-2}	2.81×10^{-2}	2.91×10^{-2}	$\mathbf{2.54\times10^{-2}}$
	ISE	5.96×10^{-2}	5.89×10^{-3}	1.81×10^{-3}	1.35×10^{-3}	1.25×10^{-3}	$\mathbf{7.86\times10^{-4}}$
	ITSE	7.13×10^{-3}	7.21×10^{-4}	3.6×10^{-4}	2.86×10^{-4}	2.12×10^{-4}	$\mathbf{2.23\times10^{-4}}$

$T_{PS1}/4$ and $4\times T_{PS1}$ are considered, and responses of ΔF_1, ΔF_2 and $\Delta Ptie$ with respect to nominal values are depicted in Fig. 9(a)–(c) respectively.

The maximum OS (0.0097) and US (0.0135) are obtained in ΔF_1 while considering $T_{PS1}/4$. The values of t_s are 2.6 s, 2.5 s, and 2.3 s are noted for ΔF_1, ΔF_2 and $\Delta Ptie$ respectively for $T_{PS1}/4$. The impact of $4\times T_{PS1}$ is negligible on the outcomes of the controller, as analyzed in Table 4. Moreover, the other uncertainties in K_{PS1} as $3\times K_{PS1}$ and $K_{PS1}/3$ are considered, and the controller performance is revealed in Fig. 10(a)–(c) and Table 4. The maximum OS (0.0083) and US (0.0092) are obtained in ΔF_1 while considering $3\times K_{PS1}$ with t_s values are 2.1 s, 3.1 s, and 2.3 s in ΔF_1, ΔF_2 and $\Delta Ptie$, respectively. The overall effect of $K_{PS1}/3$ is almost negligible, as noted in Table 4.

5.4. Stability analysis via eigenvalue approach

To enhance the dynamic indices of an AGC system, it is necessary to have an adequate amount of stability. For the purpose of stability evaluation, various methods are available, and

Table 4
Robustness analysis of proposed controller.

Proposed controller performance under parametric uncertainties with SLP of 13% in area-1												
Parameters	OS			US			t_r (s)			t_s (s)		
	ΔF_1 (Hz)	ΔF_2 (Hz)	$\Delta Ptie$ (puMW)	ΔF_1 (Hz)	ΔF_2 (Hz)	$\Delta Ptie$ (puMW)	ΔF_1	ΔF_2	$\Delta Ptie$	ΔF_1	ΔF_2	$\Delta Ptie$
Nominal K_{PS1}, T_{PS1}	0.002	0.00011	Negligible	0.0031	0.00072	0.0003	0.27	0.75	1	1.5	1.9	1
4*T_{PS1}	Negligible	Negligible	Negligible	0.00017	0.00032	0.00012	0.81	1.2	0.91	1	1.2	0.91
T_{PS1}/4	0.0097	0.00064	0.0002	0.0135	0.0023	0.00098	0.25	0.5	0.68	2.6	2.5	2.3
3*K_{PS1}	0.0083	0.00049	0.0002	0.0092	0.0018	0.00093	0.28	0.5	0.62	2.1	3.1	2.3
K_{PS1}/3	Negligible	Negligible	Negligible	0.0018	0.00039	0.00018	0.75	1.3	1.2	0.75	1.3	1.2

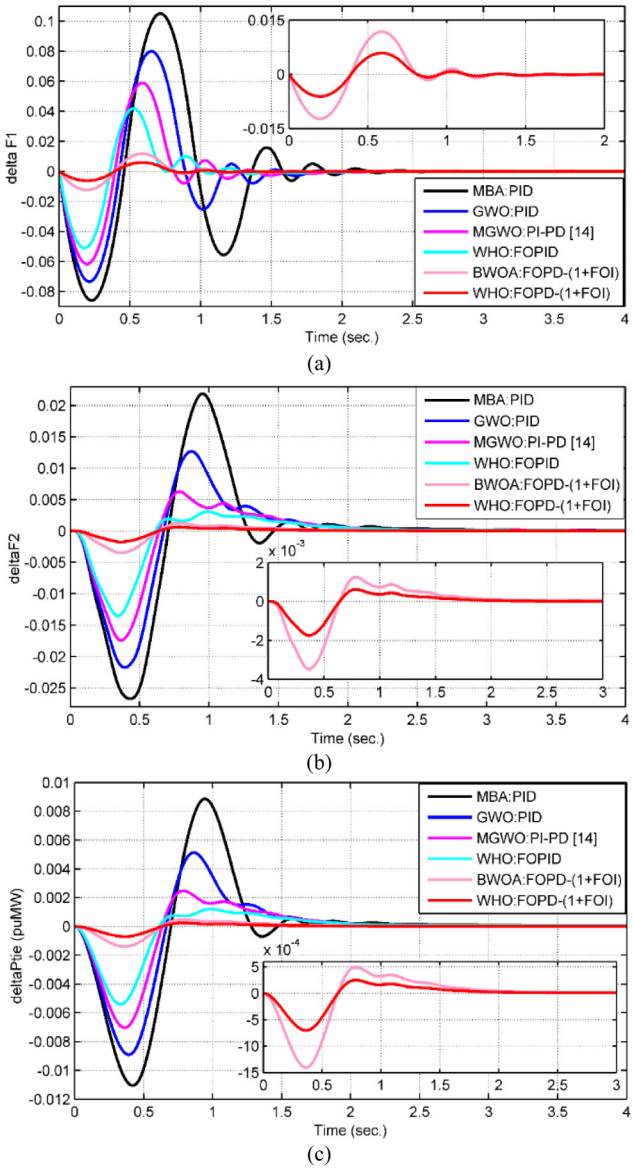


Fig. 8. Performance of controllers with SLP of 20% in Area-1 and SLP of 10% (a) ΔF_1 (Hz), (b) ΔF_2 (Hz), and (c) $\Delta Ptie$.

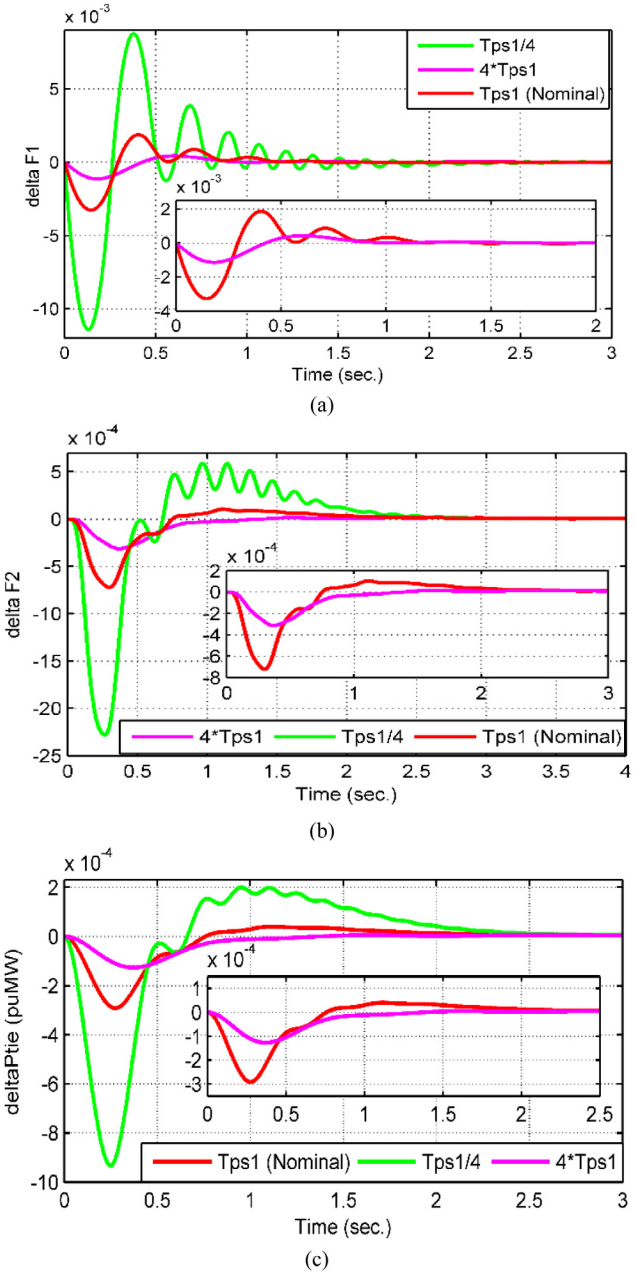


Fig. 9. Robustness analysis of proposed controller with SLP of 13% in area-1 (a) ΔF_1 (Hz) (b) ΔF_2 (Hz), and (c) $\Delta Ptie$.

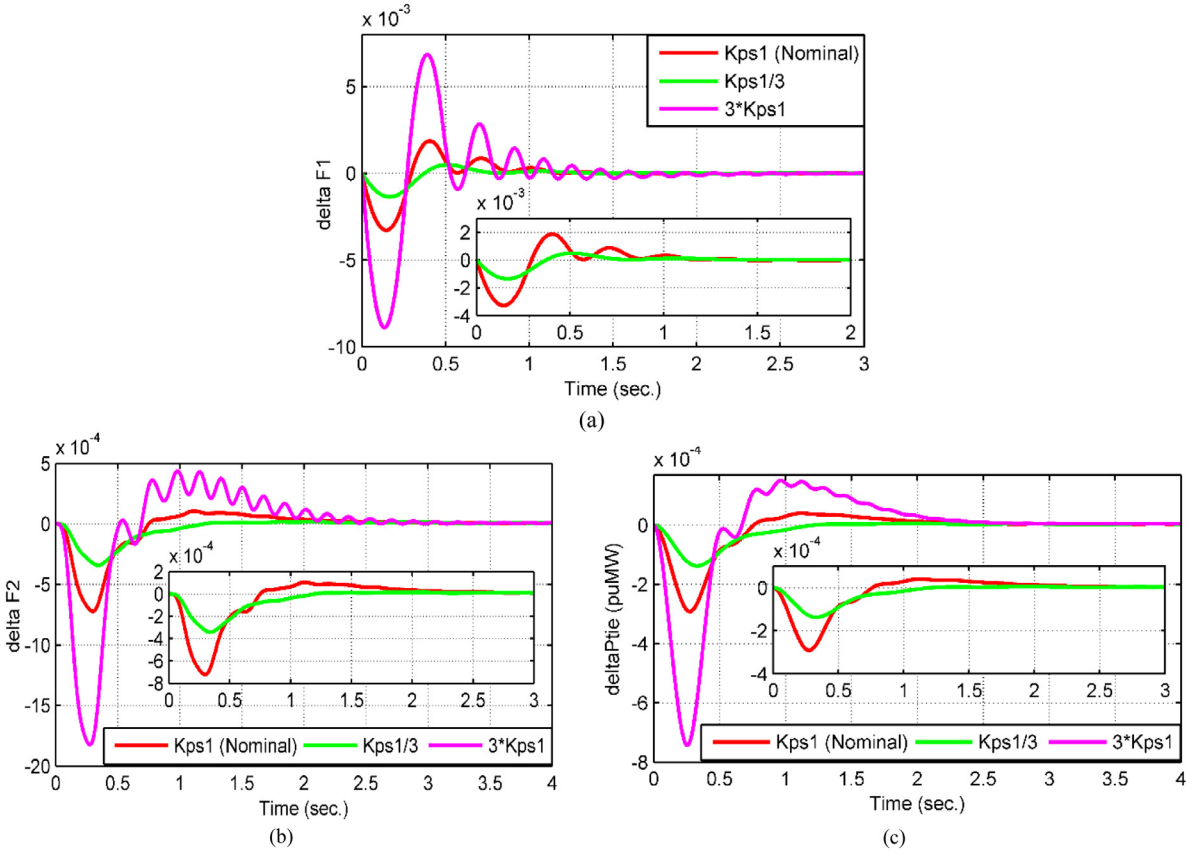


Fig. 10. Robustness analysis of proposed controller with SLP of 13% in area-1 (a) ΔF_1 (Hz) (b) ΔF_2 (Hz) (c) $\Delta Ptie$.

Table 5
Eigenvalues for area-1 and area-2.

Area-1	Area-2
0.000+0.000i	0.000+0.000i
−4.5032+47.1896i	−5.1691+43.2310i
−4.5032−47.1896i	−5.1691−43.2310i
−12.500+0.000i	−12.500+0.000i
−12.500+0.000i	−12.500−0.000i
−3.6374+0.000i	−3.7823+0.000i
−3.3333+0.000i	−3.3333+0.000i
−3.3333−0.000i	−3.3333−0.000i
−2.1923+0.000i	−2.4087+0.000i
−1.2608+0.000i	−2.0635+0.000i
−1.1381+0.000i	−1.2734+0.000i
−0.8615+0.000i	−0.3375+0.000i
−0.2001+0.000i	−0.3005+0.000i
−0.1836+0.000i	−0.1989+0.000i
−0.1531+0.000i	−0.1000+0.000i
−0.100+0.000i	−0.1000−0.000i
−0.100−0.000i	−0.0558+0.000i
−0.050+0.000i	−0.0501+0.000i
−0.050+0.000i	−0.0500+0.000i
−0.050−0.000i	−0.0500−0.000i
−0.0285+0.000i	−0.0487+0.000i
−0.0228+0.000i	−0.0072+0.000i
−0.0007+0.000i	−0.0060+0.000i

eigenvalue is one of them. Before evaluating the eigenvalues, fractional-order terms are transformed into their corresponding full-order terms via the SBL method. To show the stability of the controller, the eigenvalues of interconnected two-area PS with WHO-tuned FOPD-(1+FOI) controller are calculated and revealed in Table 5.

The TF of outer and inner loop controllers for area-1 and area-2 are revealed by Eqs. (34)–(37).

$$C_{o,1}(s) = 85.675 + 39.829s^{0.994} \cong 85.675 + 39.829s \tag{34}$$

$$C_{i,1}(s) = 1 + \frac{3.1786}{s^{0.912}} \cong 1 + \frac{3.1786}{s^{0.9}} \tag{35}$$

$$C_{o,2}(s) = 52.909 + 14.845s^{0.987} \cong 52.909 + 14.845s \tag{36}$$

$$C_{i,2}(s) = 1 + \frac{1.88296}{s^{0.1374}} \cong 1 + \frac{1.88296}{s^{0.1}} \tag{37}$$

where $s^{0.9} = \frac{242800s^4+328000s^3+60120s^2+1488s+1}{6648s^4+277300s^3+300600s^2+45610s+887.3}$ and $s^{0.1} = \frac{5532s^4+13380s^3+4147s^2+191.5s+1}{4392s^4+13530s^3+5063s^2+284.5s+1.894}$.

Table 5 depicts that all the poles of area-1 and area-2 are present in the left half of s-plane. Hence it is noted that the proposed WHO:FOPD-(1+FOI) cascade controller is able to stabilize the studied interconnected PS.

5.5. Real power system validation

To benchmark the potency of the WHO: FOPD-(1+FOI) controller, a real New England IEEE-39 test bus is considered, as shown in Fig. 11. Ten reheat-type thermal generating units are there in the standard IEEE-39 bus and segregated into three areas [3,39], as revealed in Table 6. For the validation purpose, only two areas, i.e., area-1 and area-2, are considered. The generating units of the alike area can be displaced by its equivalent single generation unit with inertia constant (H_e) and speed regulation

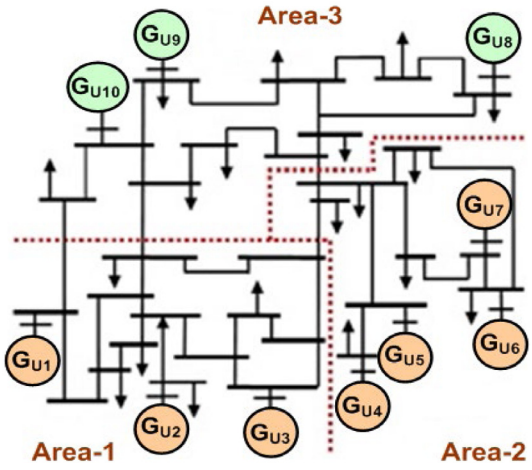


Fig. 11. New England IEEE-39 bus [3,39].

Table 6
Parameters of IEEE-39 test bus [3,5].

Area no.	Unit no.	H	R_d (Hz/MW_{pu})	D (MW_{pu}/Hz)	T_g (s)	T_t (s)
1	G_{U1}	4.0	3.0	8.33×10^{-3}	0.08	0.3
	G_{U2}	3.5	3.0	8.33×10^{-3}	0.08	0.3
	G_{U3}	2.5	1.8	8.33×10^{-3}	0.08	0.3
2	G_{U4}	2.5	2.4	8.33×10^{-3}	0.08	0.3
	G_{U5}	3.0	1.8	8.33×10^{-3}	0.08	0.3
	G_{U6}	4.0	2.1	8.33×10^{-3}	0.08	0.3
	G_{U7}	3.0	1.8	8.33×10^{-3}	0.08	0.3
$\alpha_{12} = -1, \beta_1 = \beta_2 = 0.425$ Hz/$MW_{p.u.}$, $2\pi T_{12} = 0.545$ puMW/Hz						

(R_e) as given [3,5]:

$$H_e = \frac{H_1S_1 + H_2S_2 + \ldots + H_nS_n}{S_{sys}} \tag{38}$$

$$R_e = \frac{1}{\frac{1}{R_1}\left(\frac{S_1}{S_{sys}}\right) + \frac{1}{R_2}\left(\frac{S_2}{S_{sys}}\right) + \ldots \frac{1}{R_n}\left(\frac{S_n}{S_{sys}}\right)} \text{ Hz/MW}_{pu} \tag{39}$$

$$\text{and } S_{sys} = S_1 + S_2 + \ldots + S_n \tag{40}$$

where, H_i, R_i and S_i are inertia constant, speed regulation, and machine rating of the ith generating element and G_{U1}toG_{U10} are the generator unit 1 to 10 in all three areas.

The proposed controller performance with SLP of 10% in area-1 is shown in Fig. 12. The maximum OS and US in area-1 are 0.0089 Hz and 0.010 Hz, with t_s of 1.3 s, while dealing with area-2 and tie-line flow, the maximum USs are 0.0029 Hz and 0.0011 puMW with values of t_s are 2.1 s and 2 s respectively. A vivid comparative analysis of the proposed FOPD-(1+FOI) controller on the simulated and real-time platform is revealed in Table 7.

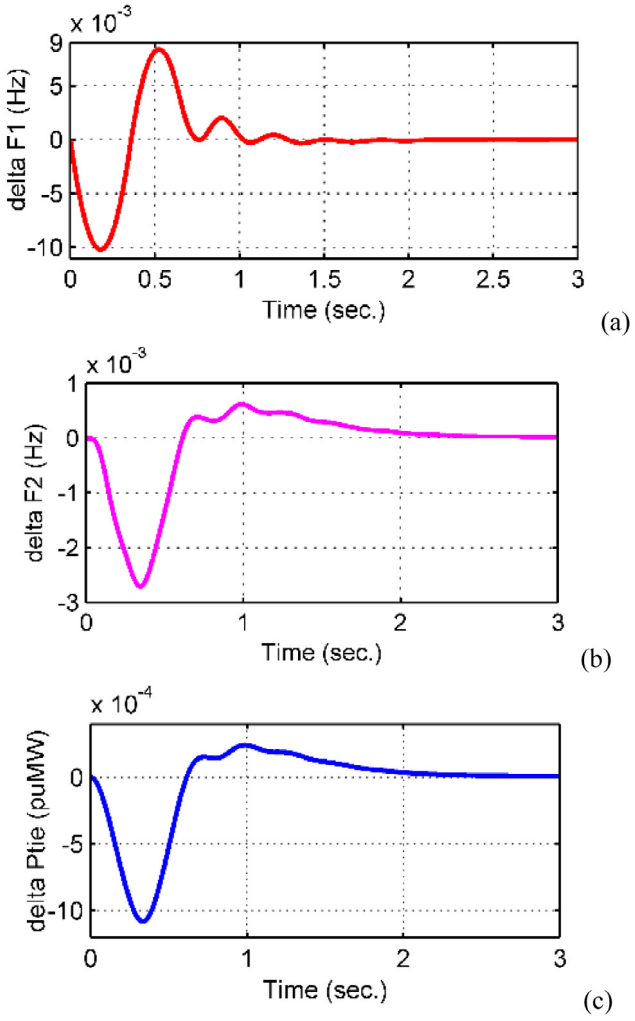


Fig. 12. Controller response for IEEE-39 bus system (a) ΔF_1 (Hz), (b) ΔF_2 (Hz), and (c) $\Delta Ptie$.

6. Conclusion

In this work first time, a novel configuration of WHO-assisted FOPD-(1+FOI) fractional cascade controller is implemented for the LFM of an interconnected reheat-type thermal PS network. In the modeling, various non-linearities, such as GRC and GDB are incorporated to replicate the practical scenarios. This controller has the feature of both a fractional-order type controller as well as a cascade-type controller. Assessment of the proposed control strategy has been analyzed over five state-of-the-art metaheuristics as MBA, GWO, MGWO, BWOA, and WHO, with four types of controllers such as PID, PI-PD, FOPID, and FOPD-(1+FOI) under-considered step load perturbations. The results obtained reveal the enhanced performance of the proposed controller in terms of best score (minimization of the objective function), computational cost, speed of convergence, transient response, steady-state response, and controller performance in terms of error. While considering the robustness evaluation under practical scenarios, the outcomes of the proposed controller are quite satisfactory with lesser OS and US, and faster settling time. The stability analysis of the proposed scheme is successfully evaluated via the eigenvalue approach. Finally, to benchmark, the performance of the controller on a real power system, an IEEE-39 New England bus is implemented successfully, and the comparative tabular analysis of simulated and real power system analysis shows satisfactory performance. The future scope of this work will predominately focus on the controller design for interconnected power systems under high renewable penetrations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 7
Comparative analysis of proposed controller.

SLP of 10% in area-1			
Indices		Simulated	Real-time
OS	ΔF_1 (Hz)	0.00230	0.00880
	ΔF_2 (Hz)	0.00024	0.00004
	$\Delta Ptie$ (puMW)	0.00010	0.00014
US	ΔF_1 (Hz)	0.00290	0.01000
	ΔF_2 (Hz)	0.00080	0.00270
	$\Delta Ptie$ (puMW)	0.00035	0.00110
Rise time (t_r) (s)	ΔF_1	0.4000	0.4100
	ΔF_2	0.6500	0.6700
	$\Delta Ptie$	0.6200	0.6700
Settling time (t_s) (s)	ΔF_1	1.3000	1.4500
	ΔF_2	1.9000	1.9200
	$\Delta Ptie$	1.9000	1.9200
ΔF_1	*IAE*	3.3×10^{-2}	4.6×10^{-2}
	ITAE	2.9×10^{-2}	4.1×10^{-2}
	ISE	1.1×10^{-3}	2.5×10^{-3}
	ITSE	3.1×10^{-4}	4.8×10^{-4}
ΔF_2	*IAE*	2.95×10^{-2}	4.45×10^{-2}
	ITAE	2.65×10^{-2}	4.31×10^{-2}
	ISE	8.21×10^{-4}	8.96×10^{-4}
	ITSE	2.3×10^{-4}	2.92×10^{-4}
$\Delta Ptie$	*IAE*	2.85×10^{-2}	3.48×10^{-2}
	ITAE	2.54×10^{-2}	2.94×10^{-2}
	ISE	7.86×10^{-4}	8.93×10^{-4}
	ITSE	2.23×10^{-4}	2.65×10^{-4}

Appendix. Parameters of 2-area reheat thermal PS [6]:

$R_i = 2.4$ Hz/puMW, $\beta_i = 0.425$ Hz/$MW_{p.u.}$, $T_{Ti} = 0.3$ s, $T_{Gi} = 0.08$ s, $T_{ri} = 10$ s, $K_{ri} = 0.5$, $K_{PSi} = 120$ Hz/puMW, $T_{PSi} = 20$ s, $\alpha_{12} = -1$, $2\pi T_{12} = 0.545$ puMW/Hz, where, i = 1, 2.

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